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Module design for pilots training with full flight simulator

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1. Background

Full flight simulator (FFS) is a sophisticated simulation system to simulate all aircraft systems that are accessible from the flight deck and are critical to training. For instance, it can simulate the force feedback for the pilot's flight controls, the avionics system, the communication system, the cockpit sounds, the aerodynamics, and the ground handling. It prepares pilots for realistic flight situations and is used for pilots training.

FFS has extremely high fidelity and is typically expensive the run. This will limit the number of sessions each pilot can try with FFS. Apart from initial training, the pilots must carry out recurrent training at regular intervals (such as every six months) in order to retain their qualification. In addition, pilots' reactions to different situations and flight skills are different. Therefore, it is important to design efficient personal training sessions for pilots to detect and improve their inadequacies.

2. Problem Formulation

We aim to design training modules with simulation-based optimization approach. Suppose there are M different training modules from which m are selected in one training session due to the time and cost restriction. Our purpose is to identify the weakness of the pilot with a carefully designed training session. The optimization problem can be formulated as follows:

$$\min_{x_1,\ldots,x_m} \mathbb{E}_\xi[\min\{Y(x_1,\xi),\ldots,Y(x_m,\xi)\}], \tag{1}$$

where $x_1,\ldots,x_m$ represent the m modules in the training session, taken from the module set $\mathcal{X} = \{x_1,\ldots,x_M\}$ and $Y(x_1,\xi)$ is the score the pilot obtain in module x_i. It is evaluated through simulation at FFS. In this model, we assume the score vector in the population $\{Y(x_1),\ldots,Y(x_M)\}$ follows a multivariate normal distribution. For each specific person, his score is a sample from this distribution. In equation (1), the random vector ξ is used to illustrate the randomness in this multivariate normal distribution.

The inner minimization is to find the minimum score one pilot obtain across the given m training modules. We take the minimum value as we need to find the weakness of the pilots considering flight safety. The outer minimization is the find the best set of training sessions in detecting the pilot inadequacy. This formulation is not for 'personal' module design, as we choose the modules that performs the best in expectation over the whole population.

This problem involves two stages. The first stage is to learn the multivariate normal distribution as well as to identify the optimal modules. This is the simulation optimization process. In the two stage, we deploy the recommended session to train the pilots.

3. Methodology

We assume that the score has Gaussian noise: $Y(x) = Z(x) + \mathcal{N}(0,\sigma(x))$ and the joint distribution for $Y(x)_{x\in\{\mathcal{X}\}}$ is a multivariate normal distribution with mean vector $\theta = [Z(x_1),\ldots,Z(x_M)]^T$ and covariance matrix Λ. We use a Bayesian approach to model θ and assume a normal prior:

$$\theta \sim \mathcal{N}(\mu_0,\Sigma_0).$$

As with Xie et al. (2016), we assume Λ, μ_0, Σ_0 are known.

The ith entry of a vector v is denoted as $v(i)$ and the (i,j)th entry of a matrix M is denoted as $M(i,j)$. For an ordered collection of m alternatives $\mathbf{x} = (x_1,\ldots,x_m)$ with element $x_i \in \{1,\ldots,M\}$ for

each i, we use $v(\mathbf{x})$ to denote the subvector of v with the ith entry equal to $v(x_i)$. We further denote by $M(\mathbf{x}, \mathbf{x}')$ the m-by-m submatrix of M with the (i, j)th entry equal to $M(x_i, x'_j)$.

We consider a situation where in each iteration n, one pilot will attend m different modules $\mathbf{x}_n = (x_{n,1}, ..., x_{n,m})^T$ and we obtain his score vector $\mathbf{y}_n = (Y(x_{n,1}), ..., Y(x_{n,m}))^T$. The conditional distribution of $\mathbf{y}_n$ is:

$$\mathbf{y}_n | \theta, \mathbf{x}_n \sim \mathcal{N}(\theta(\mathbf{x}_n), \Lambda(\mathbf{x}_n, \mathbf{x}_n)).$$

Let $\mathbb{X}_n = (\mathbf{x}_1^T, ..., \mathbf{x}_n^T)$ denote the concatenation of the design points of the previous n iteration and similarly $\mathbb{Y}_n = (\mathbf{y}_1^T, ..., \mathbf{y}_n^T)^T$. Then, the posterior distribution for θ is:

$$\theta_n | \mathbb{X}_n, \mathbb{Y}_n \sim \mathcal{N}(\mu_n, \Sigma_n),$$

where for any vector $\mathbf{x} = (x_1, x_2, ..., x_m)$,

$$\mu_n(\mathbf{x}) = \mu_0(\mathbf{x}) + \Sigma_0(\mathbf{x}, \mathbb{X}_\mathbf{n})(\mathbf{\Sigma_0}(\mathbb{X}_\mathbf{n}, \mathbb{X}_\mathbf{n}) + \mathbf{\Gamma_n})^{\mathbf{-1}}(\mathbb{Y}_\mathbf{n} - \mu_\mathbf{0}(\mathbb{X}_\mathbf{n})),$$

$$\Sigma_n(\mathbf{x}, \mathbf{x}) = \Sigma_0(\mathbf{x}, \mathbf{x}) - \Sigma_0(\mathbf{x}, \mathbb{X}_\mathbf{n})(\mathbf{\Sigma_0}(\mathbb{X}_\mathbf{n}, \mathbb{X}_\mathbf{n}) + \mathbf{\Gamma_n})^{\mathbf{-1}}\mathbf{\Sigma_0}(\mathbb{X}_\mathbf{n}, \mathbf{x}),$$

where Γ_n is the block diagonal matrix with n blocks: $\Lambda(\mathbf{x}_1, \mathbf{x}_1), ..., \Lambda(\mathbf{x}_n, \mathbf{x}_n)$.

We adopt the Expected Improvement (EI) acquisition function to select the m courses for the next iteration $n + 1$. For any candidate vector $\mathbf{x} = (x_1, ..., x_m)$,

$$\mathrm{EI}(\mathbf{x}) = \mathbb{E}_{\theta_n}[(g_c - G(\mathbf{x}))^+ | \theta_n],$$

where $(a)^+ = a$ if $a \geq 0$ and $(a)^+ = 0$ otherwise. When θ_n takes a value $\tilde{\theta_n}$, we define $G(\mathbf{x})$ as:

$$G(\mathbf{x}) = \mathbb{E}_\xi[\min\{Y(x_1, \xi), ..., Y(x_m, \xi)\}],$$

where $\{Y(x_1), ..., Y(x_m)\} \sim \mathcal{N}(\tilde{\theta_n}(\mathbf{x}), \Lambda(\mathbf{x}, \mathbf{x}))$ and $\{Y(x_1, \xi), ..., Y(x_m, \xi)\}$ is a random sample from this distribution. In the EI function, g_c is the current best value: $g_c = \min\{G(\mathbf{x}_1), ..., G(\mathbf{x}_n)\}$.

We next explain how to compute g_c. For any vector $\mathbf{x}_i, 1 \leq i \leq n$, we approximate $G(\mathbf{x}_i)$ through sample average approximation (SAA). Specifically, we generate samples of $\{Y(x_{i,1}, \xi), ..., Y(x_{i,m}, \xi)\}$ from distribution $\mathcal{N}(\mu_n(\mathbf{x}_i), \Lambda(\mathbf{x}_i, \mathbf{x}_i))$. Here, as we already have observations at $\mathbf{x}_i$, we use the posterior mean $\mu_n(\mathbf{x}_i)$ as if it were the true value of $\theta_n(\mathbf{x}_i)$ to compute the current best value. Similar approach has been widely used for ordinary stochastic GP based optimization algorithms. The samples can then be generated as follows. Suppose $\Lambda(\mathbf{x}_i, \mathbf{x}_i) = B_i B_i^T$ and $\mathbf{z}_j \in \mathbb{R}^{m \times 1}$ is a random draw from m iid standard normal distribution. The j-th random samples can

be represented as: $(Y(x_{i,1},\xi_j),...,Y(x_{i,m},\xi_j))^T = \mu_n(\mathbf{x}_i) + B_i\mathbf{z}_j$ and $Y(x_{i,k},\xi_j) = \mu_n(x_{i,k}) + B_i^k\mathbf{z}_j$, where B_i^k is the kth row of B_i. Therefore, we have

$$G(\mathbf{x}_i) \approx \frac{1}{J}\sum_{j=1}^{J}\min\mu_n(\mathbf{x}_i) + B_i\mathbf{z}_j = \frac{1}{J}\sum_{j=1}^{J}\min\{\mu_n(x_{i,1}) + B_i^1\mathbf{z}_j,...,\mu_n(x_{i,m}) + B_i^m\mathbf{z}_j\}.$$

To compute EI for $\mathbf{x}$, we should further take care of the outer expectation with respect to the posterior distribution of θ. We can use similar approach as above the generate samples from the posterior distribution of $\theta_n(\mathbf{x})$ and obtain the following SAA form:

$$\mathrm{EI}(\mathbf{x}) \approx \frac{1}{K}\sum_{k=1}^{K}(g_c - \frac{1}{J}\sum_{j=1}^{J}\min\theta_n(\mathbf{x}) + A\tilde{\mathbf{z}}_k + B\mathbf{z}_j)^+,$$

where $\Sigma_n(\mathbf{x},\mathbf{x}) = AA^T$, $\Lambda(\mathbf{x},\mathbf{x}) = BB^T$, and both $\tilde{\mathbf{z}}_k$ and $\mathbf{z}_j$ are iid standard normal vectors of length m.

4. Some special cases

4.1. Two normal variables

We need to compare $\mathrm{E}_{\xi_1,\xi_2}[\min\{\mu_1 + A\xi_1, \mu_2 + B_1\xi_1 + B_2\xi_2\}]$ and $\mathrm{E}_{\xi_1,\xi_2}[\min\{\mu_1 + A\xi_1, \mu_2 + B\xi_2\}]$, where ξ_1,ξ_2 are independent standard normal random variable. All the coefficients are positive and that $B_1^2 + B_2^2 = B^2$.

Notice that for any two Gaussian random number $X_1 \sim N(\mu_1,\sigma_1^2)$, $X_2 \sim N(\mu_2,\sigma_2^2)$ with correlation ρ, the expectation $\mathrm{E}[\min\{X_1,X_2\}]$ takes the following form (Clark 1961):

$$\mathrm{E}[\min\{X_1,X_2\}] = \mu_1\Phi(\frac{\mu_2-\mu_1}{\theta}) + \mu_2\Phi(\frac{\mu_1-\mu_2}{\theta}) - \theta\phi(\frac{\mu_2-\mu_1}{\theta}),$$

where ϕ, Φ are cdf and pdf for standard normal distribution, respectively, and $\theta = \sqrt{\sigma_1^2+\sigma_2^2-2\rho\sigma_1\sigma_2} = \sqrt{\mathrm{var}(X_1-X_2)}$.

We define $X_1 = \mu_1 + A\xi_1 \sim N(\mu_1,A^2)$, $X_2 = \mu_2 + B_1\xi_1 + B_2\xi_2 \sim N(\mu_2,B_1^2+B_2^2)$, $X_3 = \mu_2 + B\xi_2 \sim N(\mu_2,B^2)$. Thus,

$$\mathrm{cov}(X_1,X_2) = AB_1, \mathrm{cov}(X_1,X_3) = 0.$$

$$\mathrm{var}(X_1-X_2) = A^2 + B_1^2 + B_2^2 - 2AB_1, \mathrm{var}(X_1-X_3) = A^2 + B^2.$$

Therefore,

$$\mathrm{E}[\min\{X_1,X_2\}] = \mu_1\Phi(\frac{\mu_2-\mu_1}{\theta_1}) + \mu_2\Phi(\frac{\mu_1-\mu_2}{\theta_1}) - \theta_1\phi(\frac{\mu_2-\mu_1}{\theta_1}),$$
$$\mathrm{E}[\min\{X_1,X_3\}] = \mu_1\Phi(\frac{\mu_2-\mu_1}{\theta_2}) + \mu_2\Phi(\frac{\mu_1-\mu_2}{\theta_2}) - \theta_2\phi(\frac{\mu_2-\mu_1}{\theta_2}),$$

where $\theta_1 = \sqrt{A^2 + B_1^2 + B_2^2 - 2AB_1}$, $\theta_2 = \sqrt{A^2 + B^2}$. Denote:

$$f(\theta) = \mu_1\Phi(\frac{\mu_2 - \mu_1}{\theta}) + \mu_2\Phi(\frac{\mu_1 - \mu_2}{\theta}) - \theta\phi(\frac{\mu_2 - \mu_1}{\theta}) = (\mu_1 - \mu_2)\Phi(\frac{\mu_2 - \mu_1}{\theta}) + \mu_2 - \theta\phi(\frac{\mu_2 - \mu_1}{\theta}).$$

We have

$$\begin{aligned} f'(\theta) &= (\mu_1 - \mu_2)\phi(\frac{\mu_2 - \mu_1}{\theta})\frac{\mu_2 - \mu_1}{-\theta^2} - \phi(\frac{\mu_2 - \mu_1}{\theta}) - \theta\phi(\frac{\mu_2 - \mu_1}{\theta})\frac{\mu_2 - \mu_1}{-\theta}\frac{\mu_2 - \mu_1}{-\theta^2} \\ &= -\phi(\frac{\mu_2 - \mu_1}{\theta}) < 0 \end{aligned}$$

Therefore, f_θ is a decreasing function w.r.t. θ. We have the following conclusions:

1. When $B_1^2 + B_2^2 = B^2$ and all coefficients are positive, we have $\theta_1 = \sqrt{A^2 + B^2 - 2AB_1} < \theta_2$. Hence, $\mathrm{E}[\min\{X_1, X_2\}] > \mathrm{E}[\min\{X_1, X_3\}]$.

2. When $B_1^2 + B_2^2 = B^2$ and $B_1 < 0$, we have $\theta_1 = \sqrt{A^2 + B^2 - 2AB_1} > \theta_2$. Hence, $\mathrm{E}[\min\{X_1, X_2\}] < \mathrm{E}[\min\{X_1, X_3\}]$.

Another way of presenting these results. Suppose $X_1 \sim N(\mu_1, \sigma_1^2)$, $X_2 \sim N(\mu_2, \sigma_2^2)$ with correlation ρ and $\theta = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} = \sqrt{\mathrm{var}(X_1 - X_2)}$. Denote $f = \mathrm{E}[\min\{X_1, X_2\}]$. We have:

$$\frac{\partial f}{\partial \theta} = -\phi(\frac{\mu_2 - \mu_1}{\theta}) < 0, \tag{2}$$

$$\frac{\partial f}{\partial \rho} = \frac{\partial f}{\partial \theta}\frac{\partial \theta}{\partial \rho} = \frac{\sigma_1\sigma_2}{\theta}\phi(\frac{\mu_2 - \mu_1}{\theta}) > 0, \tag{3}$$

$$\frac{\partial f}{\partial \sigma_1} = \frac{\partial f}{\partial \theta}\frac{\partial \theta}{\partial \sigma_1} = -\frac{\sigma_1 - \rho\sigma_2}{\theta}\phi(\frac{\mu_2 - \mu_1}{\theta}), \tag{4}$$

$$\frac{\partial f}{\partial \mu_1} = \Phi(\frac{\mu_2 - \mu_1}{\theta}) > 0. \tag{5}$$

We summarize the conclusions as follows:

1. From (3) and (5), we observe that variables with smaller correlations and expectations are preferred.

2. The relation between the objective function and the variance for each individual variable is not monotone. We can, however, derive some special cases:

(a) When $\rho < \frac{\sigma_1}{\sigma_2}$ (specifically when $\rho < 0$), a larger value of σ_1 is preferred.

(b) When $\rho > \frac{\sigma_1}{\sigma_2}$, smaller value of σ_1 is preferred. This seems a little bit counter-intuitive. We can consider some special situation to understand this. Consider when $\rho = 1$ and $\sigma_1 < \sigma_2$. In this situation, X_1 and X_2 are positively linear dependent: $X_1 = \frac{\sigma_1}{\sigma_2}X_2$. We can find that they always have the same sign and when X_1 and $X_2 > 0$, $\min\{X_1, X_2\} = X_1 > 0$, which increases with σ_1. When

X_1 and $X_2 < 0$, $\min\{X_1, X_2\} = X_2 < 0$, which does not change with σ_1. Therefore, in this case, a larger value of σ_1 will increase the objective value. Although a larger value of σ_1 will reduce X_1 when it is negative, these negative values, however, are not counted in the objective function as long as $\sigma_1 < \sigma_2$. This situation will slightly gets better when ρ deviates from 1, but still makes less negative values of X_1 to be counted in the objective function until $\rho < \frac{\sigma_1}{\sigma_2}$.

4.2. Several normal variables

We consider the following situations: m normal random numbers $X_1, ..., X_m$ with equal mean μ and equal variance σ^2. The correlation between each pair of variables is $\rho \geq 0$. We would like to explore the relationship between the objective function $f = \mathrm{E}[\min\{X_1, ..., X_m\}]$ and ρ, μ, and σ.

In this case, each random variable can be represented as $X_i = \mu + \sigma(\sqrt{\rho}W + \sqrt{1-\rho}N_i)$, where W and $N_1, ..., N_m$ are i.i.d. standard normal random variables. Then,

$$f = \mathrm{E}[\min\{X_1, ..., X_m\}] = \mathrm{E}[\min\{\mu + \sigma(\sqrt{\rho}W + \sqrt{1-\rho}N_1), ..., \mu + \sigma(\sqrt{\rho}W + \sqrt{1-\rho}N_m)\}]$$

$$= \mu + \sigma\mathrm{E}[\min\{\sqrt{\rho}W + \sqrt{1-\rho}N_1, ..., \sqrt{\rho}W + \sqrt{1-\rho}N_m\}]$$

$$= \mu + \sigma\mathrm{E}[\sqrt{\rho}W + \sqrt{1-\rho}\min\{N_1, ..., N_m\}] = \mu + \sigma\sqrt{1-\rho}\mathrm{E}[\min\{N_1, ..., N_m\}].$$

This had no analytical solution. Some special cases:

1. When $\rho = 1$, $f = \mu$.
2. When $\rho = 0$, $f = \mu + \sigma\mathrm{E}[\min\{N_1, ..., N_m\}]$.

$\mathrm{E}[\min\{N_1, ..., N_m\}]$ is a constant for a given m. It is negative and can be approximated as

$$-\sqrt{2\ln m} + \frac{\ln\ln m + \ln 4\pi}{2\sqrt{2\ln m}}.$$

Hence, $f \approx \mu - \sigma\sqrt{1-\rho}\sqrt{2\ln m}$.

We can see that in this case, smaller μ, larger σ and smaller ρ are preferred. Intuitively, when ρ decreases, the m variables are less correlated and thus each of them has larger probability to take smaller values more independently. Therefore, the objective value increases as ρ.

5. Simulations with known θ and Λ

5.1. SAA approach

We first show that how we get the best modules $\mathbf{x} = (x_1, ..., x_m)$ if we already known θ and Λ. Suppose we now need to select 8 out of 50 modules such that the expected minimum value of their

scores is minimized. In connection with what we said above about SAA, one of the most natural ideas is to take a submodule approach to utilizing SAA to select modules sequentially. The steps are as follows:

1. Let $x_1 = \arg\min_{x\in\mathcal{X}} \theta(x)$, get the collection of selected modules $S = \{x_1\}$
2. Let $x_2 = \arg\min_{x\in\mathcal{X}/\{x_1\}} \mathrm{E}[\min\{x_1, x\}]$, update the collection $S = \{x_1, x_2\}$
3. Let $x_3 = \arg\min_{x\in\mathcal{X}/S} \mathrm{E}[\min\{x_1, x_2, x\}]$, update the collection $S = \{x_1, x_2, x_3\}$
4. ... keep doing this until we get the collection $S = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8\}$
5. Output $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8)$ as the result

In the above steps we frequently use SAA to approximate the minimum expectation of modules. Considering that SAA consumes large arithmetic resources, this may affect the application of the method in large-scale problems. Therefore in the next subsection we will show another approximation of the computation, which has much less computational complexity compared to SAA.

5.2. Improvement-based approach

We define the improvement of module x_2 over module x_1 as $\mathrm{Ipr}(x_2, x_1)$:

$$\mathrm{Ipr}(x_2, x_1) = \mathrm{E}[\min\{x_1, x_2\}] - \mathrm{E}[x_1]$$

Obviously, when x_1 and x_2 are exactly the same, $\mathrm{Ipr}(x_2, x_1) = 0$. Next, suppose we have a set of modules $S = \{x_1, x_3, x_4\}$, we can samely define the improvement of module x_2 over set S as $\mathrm{Ipr}(x_2, S)$:

$$\mathrm{Ipr}(x_2, S) = \mathrm{E}[\min\{S, x_2\}] - \mathrm{E}[\min\{S\}]$$

Now let's consider an extreme case, assuming x_1 and x_2 are exactly the same (i.e. $\mathrm{Ipr}(x_2, x_1) = 0$), regardless of the value of $\mathrm{Ipr}(x_2, x_3)$ and $\mathrm{Ipr}(x_2, x_4)$, the value of $\mathrm{Ipr}(x_2, S)$ is always 0. This inspires us to speculate that the value of the improvement of module x_2 over set S mainly depends on $\min_{x\in S} \mathrm{Ipr}(x_2, x)$. Based on this we can get our improvement-based approach as follows:

1. Let $x_1 = \arg\min_{x\in\mathcal{X}} \theta(x)$, get the collection of selected modules $S = \{x_1\}$
2. Let $x_2 = \arg\min_{x\in\mathcal{X}/\{x_1\}} \mathrm{E}[\min\{x_1, x\}]$, update the collection $S = \{x_1, x_2\}$ (Step 1.and 2. are exactly the same as SAA approach)
3. Let $x_3 = \arg\max_{x\in\mathcal{X}/S} \left(\min_{y\in S} \mathrm{Ipr}(x, y)\right)$, update the collection $S = \{x_1, x_2, x_3\}$
4. Let $x_4 = \arg\max_{x\in\mathcal{X}/S} \left(\min_{y\in S} \mathrm{Ipr}(x, y)\right)$, update the collection $S = \{x_1, x_2, x_3, x_4\}$
5. ... keep doing this until we get the collection $S = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8\}$

6. Output $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8)$ as the result.

The $\min_{y \in S} \mathrm{Ipr}(x, y)$ in the above process can be quickly calculated using the formula in 4.1, which saves a lot of computing resources compared to SAA.

5.3. Mixed approach

Although Improvement-based approach has extremely high computational efficiency, there may be a certain gap in the quality of its output results compared to SAA approach. Therefore, we are considering combining the advantages of both. At each step of the submodule process, we first use Improvement-based approach to preliminarily screen all modules, eliminating a certain number of modules to reduce the number of candidate modules. Then use SAA approach to select from the remaining modules. The specific process is as follows:

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